

1.3 Chemical Bonding

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.3 Chemical Bonding
Difficulty	Easy

Time allowed: 20

Score: /12

Percentage: /100

Question 1

Phosphine, PH_3 , can react with a hydrogen ion, H^+ , to form the phosphonium ion.

Which type of bond is formed in this reaction?

- A. Dipole-dipole forces
- B. Dative covalent bond
- C. Ionic bond
- D. Hydrogen bond

[1 mark]

Question 2

Silver and iodine are both shiny crystalline solids.

Which forces exist between neighboring iodine molecules in solid iodine and particles in solid silver?

	Iodine	Silver
A	Metallic bonds	Covalent bonds
B	Ionic bonds	Metallic
C	Covalent bonds	Covalent bonds
D	Van der Waals' forces	Metallic

[1 mark]

Question 3

Below are four solids. Which of these contains more than one kind of bonding?

- A. Diamond
- B. Sodium chloride
- C. Iron
- D. Ice

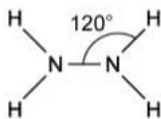
[1 mark]

Question 4

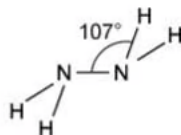
Hydrazine, N_2H_4 , is a precursor for multiple pharmaceutical compounds.

Which of the diagrams below illustrate the most likely structure and bond angle of hydrazine?

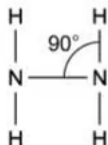
A.



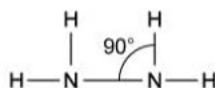
B.



C.



D.



[1 mark]

Question 5

Which of the following dipole labels are not correct?

- A. $\delta^+ \delta^-$ in propanol
 $\text{C}-\text{O}$
- B. $\delta^+ \delta^-$ in propanal
 $\text{C}=\text{O}$
- C. $\delta^+ \delta^-$ in propane
 $\text{C}-\text{H}$
- D. $\delta^+ \delta^-$ in chloropropane
 $\text{C}-\text{Cl}$

[1 mark]

Question 6

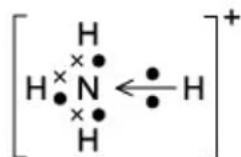
In acidic conditions, ammonia, NH_3 , reacts with a proton, H^+ , to form ammonium NH_4^+ .

Using the following key:

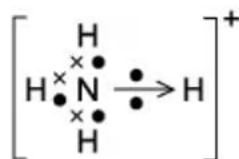
- N electron
- × H electron

Which of the following Dot & Cross diagrams correctly illustrate electron movement in this reaction?

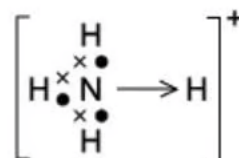
A.



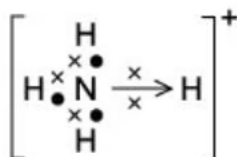
B.



C.



D.



[1 mark]

Question 7

Two boron-containing components, boron trifluoride (BF_3) and sodium borohydride (NaBH_4), have different molecular shapes.

What are the shapes around the boron atom in these molecules?

	Boron trifluoride (BF_3)	Borohydride ion (BH_4^-)
A	Pyramidal	Tetrahedral
B	Pyramidal	Square planar
C	Trigonal planar	Tetrahedral
D	Trigonal planar	Square planar

[1 mark]

Question 8

Which of these atoms is most electronegative?

- A. Cl
- B. Mg
- C. Br
- D. Na

[1 mark]

Question 9

Which of the following compounds is not bonded ionically?

- A. CaCO_3
- B. CH_3OH
- C. NaOH
- D. BaCl_2

[1 mark]

Question 10

“Electrostatic attraction between cations and delocalised electrons”

Which of the following bonds does the statement best describe?

- A. Hydrogen
- B. Ionic
- C. Dipole
- D. Metallic

[1 mark]

Question 11

Which of the following materials only contain one type of bond in solid form?

- A. Graphite
- B. Brass (an alloy of zinc and copper)
- C. Ice
- D. Iodine crystals

[1 mark]

Question 12

Which statement correctly describes what happens when silicon tetrachloride is added to water?

- A. The SiCl_4 reacts to give an acidic solution only
- B. The SiCl_4 reacts to give a precipitate and an acidic solution
- C. The SiCl_4 reacts to give a precipitate and a neutral solution
- D. The SiCl_4 dissolves to give a neutral solution only

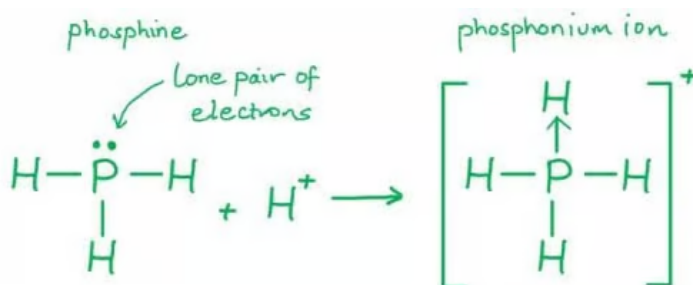
[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- A dative or coordinate bond is a covalent bond (a shared pair of electrons) in which both electrons come from the same atom.
- The H^+ ion has no electrons to share, so the lone pair of electrons make the dative covalent bond.



A is incorrect as dipole-dipole forces result when two dipolar molecules interact with each other.

C is incorrect as ionic bonding is the formation of a bond by the complete transfer of valence electrons between atoms.

D is incorrect as hydrogen bonds are a strong type of dipole interaction between a hydrogen atom attached to a highly electronegative atom and a second molecule with a lone pair of electrons on a highly electronegative atom.

2

The correct answer is **D** because:

- Van der Waals interactions are the weakest of all intermolecular attractions between molecules.
- There are three types of Van der Waals forces; dispersion, dipole-dipole, and

hydrogen bonding.

- Dispersion (or London forces) are temporary or induced dipoles in molecules that attract each other weakly, the molecular size and shape will affect the strength of these forces and the boiling points of the molecules.
 - In between iodine molecules there are strong instantaneous dipoles, a type of dispersion.
- Metallic bonding is formed between metal atoms and share the valence electrons in a sea of delocalised electrons.
- The metal is held together by these strong forces of attraction between the positive nuclei and the delocalised electrons.
 - Silver is a metal so the bond between the silver atoms will be metallic.

A is incorrect as silver is a metal and iodine is a non-metal.

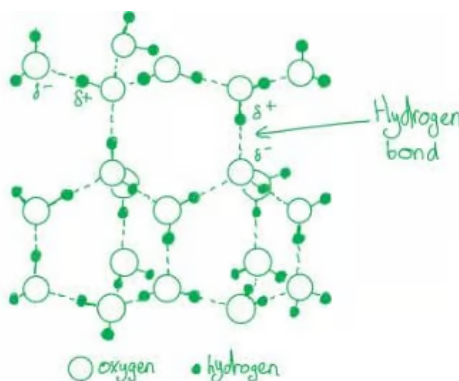
B is incorrect as iodine will not form ionic bonds; the atoms are sharing electrons.

C is incorrect as van der Waals forces are weak, short-range electrostatic attractive forces between uncharged molecules, from permanent or temporary dipoles.

3

The correct answer is **D** because:

- Ice contains both covalent bonds between the atoms in the water molecule and hydrogen bonds between the molecules.
- Hydrogen bonds are a type of dipole interaction between molecules.
- It results from the attractive force between a hydrogen atom covalently bonded to a very electronegative atom such as N, O or F.
- When hydrogen is bonded to a very electronegative atom this creates a highly polar covalent bond, hydrogen δ^+ and the other atom being δ^- .



A is incorrect as diamond has only covalent bonds.

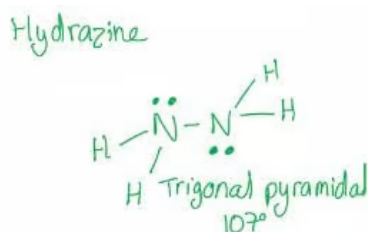
B is incorrect as sodium chloride has only ionic bonds.

C is incorrect as iron has only metallic bonds.

4

The correct answer is **B** because:

- The nitrogen atoms in hydrazine have 3 bonds (N-N and two N-H) and one lone pair to give 4 regions of electron density.
- This would give a tetrahedral shape (standard bond angle of 109.5°).
- There is a lone pair of electrons on each nitrogen atom.
- Due to the repulsion of the lone pair, the bond angle is reduced by 2.5° to 107° .



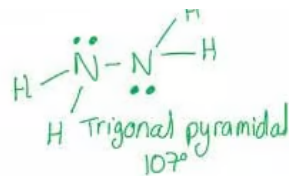
5

The correct answer is **C** because:

- the C-H bond is non-polar as there is little difference in electronegativity between

C and H.

A is incorrect as oxygen has a greater nuclear charge than carbon; it tends to polarise



5

The correct answer is **C** because:

- the C-H bond is non-polar as there is little difference in electronegativity between C and H.

A is incorrect as oxygen has a greater nuclear charge than carbon; it tends to polarise the electron density of the C-O bond to give $\delta^+C-O\delta^-$.

B is incorrect as oxygen has a greater nuclear charge than carbon; it tends to polarise the electron density of the C=O bond to give $\delta^+C=O\delta^-$.

D is incorrect as chlorine has a greater nuclear charge than carbon; it tends to polarise the electron density of the C-Cl bond to give $\delta^+C-Cl\delta^-$.

6

The correct answer is **B** because:

- The H⁺ ion has no electrons to take part in the bonding with the ammonia molecule.
- The nitrogen atom has a lone pair of electrons that will form a dative (coordinate covalent) bond.
- A dative (coordinate covalent) bond is one where both electrons come from the same molecule.

A is incorrect as the arrow is showing nitrogen electrons moving from the hydrogen ion.

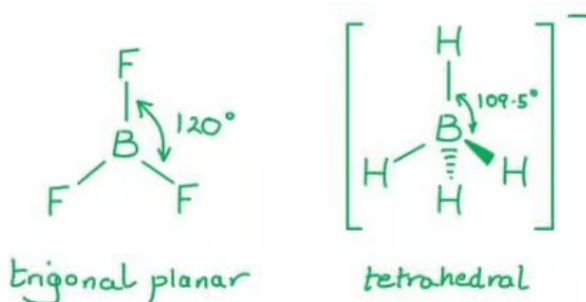
C is incorrect as there are no electrons shown.

D is incorrect as the diagram is showing hydrogen's electrons moving from the nitrogen atom to the hydrogen ion.

7

The correct answer is **C** because:

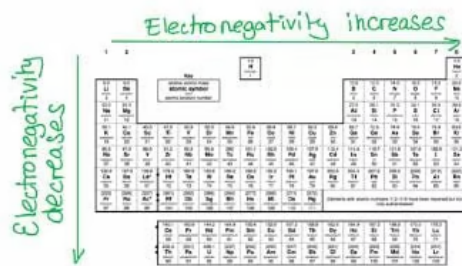
- In boron trifluoride, there are 3 atoms around the central boron atom giving equal angles of 120° and a trigonal planar structure.
- In borohydride ion, there are 4 hydrogen atoms around the central boron atom giving equal angles of 109.5° and a tetrahedral structure.



8

The correct answer is **A** because:

- The electronegativity of an atom is affected by the atomic number and the distance between the valence electrons and the nucleus.
 - Atoms with high electronegativity attract electrons strongly.
 - Atoms with low electronegativity attract electrons weakly.
- The Pauling electronegativity scale is based on the bond-energy calculations for different electron configurations.
- The electronegativity increases as you move to the right of the periodic table.
- The electronegativity decreases as you move down the periods.
- The electronegativity of chlorine is 3.16.



B is incorrect as magnesium is on the left-hand side of the periodic table so will be lower in electronegativity (1.31).

C is incorrect as bromine is below chlorine in the group so the electronegativity will be less (2.96).

D is incorrect as sodium is on the left-hand side of the periodic table so will be lower in electronegativity (0.93).

9

The correct answer is **B** because:

- Ionic bonding is the complete transfer of valence electrons between atoms.
- It creates a positively and negatively charged atom that is bonded together by the electrostatic forces of attraction between these.
- Ionic bonds generally happen between metals (losing electrons) and non-metals (gaining electrons).
- CH_3OH contains all non-metals and will, therefore, be bonded with covalent bonds.

A is incorrect as calcium is a metal, and the carbonate is a non-metal ion.

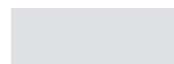
C is incorrect as sodium is a metal, and the hydroxide is a non-metal ion.

D is incorrect as barium is a metal and chlorine is a non-metal.

10

The correct answer is **D** because:

- Metallic bonding results when the electrons in the metal atoms are delocalised, and a positive ion (cation) remains.



lower electronegativity (1.07).

C is incorrect as bromine is below chlorine in the group so the electronegativity will be less (2.96).

D is incorrect as sodium is on the left-hand side of the periodic table so will be lower in electronegativity (0.93).

9

The correct answer is **B** because:

- Ionic bonding is the complete transfer of valence electrons between atoms.
- It creates a positively and negatively charged atom that is bonded together by the electrostatic forces of attraction between these.
- Ionic bonds generally happen between metals (losing electrons) and non-metals (gaining electrons).
- CH_3OH contains all non-metals and will, therefore, be bonded with covalent bonds.

A is incorrect as calcium is a metal, and the carbonate is a non-metal ion.

C is incorrect as sodium is a metal, and the hydroxide is a non-metal ion.

D is incorrect as barium is a metal and chlorine is a non-metal.

10

The correct answer is **D** because:

- Metallic bonding results when the electrons in the metal atoms are delocalised, and a positive ion (cation) remains.
- It is the attraction between the positive ion and the delocalised electrons that creates the metallic bond.

A is incorrect as a hydrogen bond is an intermolecular bond interaction between hydrogen attached to a highly electronegative element (creating a δ^+ hydrogen), with an atom with a lone pair in another molecule.

B is incorrect as ionic bonding is the complete transfer of valence electrons between atoms.

C is incorrect as a dipole is created by unequal sharing of electrons between atoms in a molecule forming regions of δ^+ and δ^- that are attracted to each other.

11

The correct answer is **B** because:

- Metallic bonding results when the electrons in the metal atoms are delocalised, and a positive ion (cation) remains.
- It is the attraction between the positive ion and the delocalised electrons that creates the metallic bond.
- The only bonding present in brass will be metallic bonding.
- This is because both atoms present are metals, zinc and copper, and will undergo the same type of bonding with each other.

A is incorrect as graphite has both covalent bonds between the carbon atoms and van de Waals forces between layer in its structure.

C is incorrect as ice will have covalent and hydrogen bonds in its structure.

D is incorrect as iodine has covalent bonds between the atoms and strong instantaneous dipole forces between the molecules.

12

The correct answer is **B** because:

- Silicon tetrachloride reacts readily with water to produce a precipitate of silicon dioxide and hydrochloric acid:
 - $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$

A is incorrect as the precipitate of silicon dioxide is produced.

C is incorrect as hydrochloric acid is produced making the product acidic.

D is incorrect as both a precipitate and an acidic solution are produced.